

Real-world (conflicting requirements design) — Hints

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Only 2 hints — work through them in order against your own written design before reading ahead. Each hint has 2 depth levels: **Basic** (the plain framing) and **Intermediate** (the real architectural vocabulary and trade-offs, including the senior-level judgment call). Read Basic first even if you already know the vocabulary — it's the fastest way to see exactly what each deeper level adds.

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Hint 1 — What to ask, and the obvious first (wrong) instinct

Basic Version

Before designing anything, find every place the requirements actually disagree — write them down as a list, plainly, one line each. For each conflict, ask: who actually has the authority to decide when these two people want different things? And what's the real risk on each side if you get the call wrong?

The tempting first instinct is to try to satisfy everyone by building the most cautious version of every single request at once — but if two requests genuinely contradict each other ("fully automatic and instant" and "a human reviews everything"), you can't build both fully. That instinct isn't a design, it's avoiding the decision.

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Intermediate Version

Name the conflict explicitly, in writing, before you design a single agent: "fast and fully autonomous" and "a human reviews everything" cannot both be fully true at once. A reasonable way to reconcile them without more information is a **tiered-risk routing pattern** — a conditional routing pattern, the same family as Doc11's router patterns: low-risk work (routine topics, well-sourced, high fact-check confidence) auto-publishes; higher-risk work (contested topics, anything touching legal/financial/medical claims, low fact-check confidence) routes to a human reviewer.

This isn't "solving" the disagreement on your own authority — it's a specific *proposed* compromise design, and it has to be flagged explicitly as an assumption, with the actual question you'd ask stakeholders written down next to it: "where should the auto-publish / requires-review line sit, and who signs off on that line?"

What a senior reviewer pushes on:

The trap at this level is over-engineering to "please everyone" — bolting on every stakeholder's individual ask (a full audit trail for legal, an autonomous fast-path for the speed request, a hard cost ceiling for finance, a manual override switch for every edge case anyone can imagine) produces something so complex nobody can explain or maintain it — and it still hasn't actually resolved the underlying disagreement about who owns the risk-tier decision. It just buries the disagreement inside code where nobody will notice it wasn't actually settled.

The senior move is the opposite of "build everything": design the smallest system that resolves the conflict through one explicit, named decision (here, the risk-tier cutoff), present that decision plainly as a stated assumption, and ask the actual stakeholders to confirm or correct it — instead of silently encoding your own guess as if it were already agreed policy. If you're pushed on why you didn't just build in every stakeholder's individual request, the answer is that unreviewed, unauthorized policy calls baked into code are exactly the kind of hidden decision a real system-design review exists to catch.

Difference between Basic and Intermediate: Basic is the discipline of writing conflicts down instead of designing around them silently, and refusing to "just build everything so nobody's upset." Intermediate supplies the real pattern (tiered-risk conditional routing), names the compromise as a compromise that needs real sign-off, and resists hiding an unresolved policy disagreement inside ever-more-complex code.

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Hint 2 — The plan, and almost the whole design

Basic Version

list every place the requirements disagree, plainly, one line each

for each conflict:

 who decides when these two people want different things?

 what's the real risk on each side if the guess is wrong?

do not: silently build the "safest" version of every request at once

 -- that avoids the decision instead of making it

design around your best guess, and write the guess down as an
 assumption, not a fact

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Intermediate Version

Step 1 -- name every conflict explicitly (example shape):

"fast, fully autonomous" (stakeholder A)

vs.

"human review of everything, no exceptions" (stakeholder B)

vs.

"minimize API/compute cost" (stakeholder C)

Step 2 -- propose a reconciling pattern, not a guess buried in code:

tiered-risk conditional routing (Doc11 routing family)

risk score = f(topic sensitivity, source confidence, fact-check score)

low risk -> auto-publish (serves "fast")

high risk -> human review (serves "safe")

Step 3 -- flag it, don't silently decide it:

write the proposed risk-tier cutoff down as a stated assumption

write the exact question for stakeholders: "where should this line sit, and who owns moving it?"

Step 4 -- address the requirement nobody reconciled yet (cost):

add a cost cap/guardrail as its own explicit piece, not folded silently into the routing logic

Hardened for a senior review:

Same 4 steps, plus the discipline of resisting "build everything":

For every stakeholder ask, before adding a component for it, ask:

does this genuinely serve a real requirement, or is it here so I can say "I addressed everyone's concern" without actually deciding anything?

Concretely reject, on purpose, and say so out loud in the design:

- a manual override switch for every conceivable edge case (this doesn't resolve the risk-tier question, it just adds a lever nobody has agreed how to use)
- a fully separate approval workflow bolted on per-stakeholder instead of one shared risk-tier decision everyone routes through

Present the design as:

1. here are the conflicts, named plainly
2. here is the proposed resolution (tiered routing) and exactly which assumption it rests on
3. here is the one open question stakeholders need to actually answer before this ships as policy, not just as code

Difference between Basic and Intermediate: Basic is the plan anyone can follow — list conflicts, ask who decides, don't dodge the decision by building everything. Intermediate turns that into a real architecture (tiered-risk routing) with a written stakeholder question, and adds the discipline a senior reviewer watches for: rejecting components that exist only to *look* like every concern was addressed.

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